

# Architectures of problem solving

(working notes for discussion)

July 2026

## Where we started

- Agents on a fixed directed network share a drifting Boolean-satisfiability task: private assignments  $x^{(i)} \in \{0, 1\}^K$ , a hidden clause set  $\mathcal{C}$  ( $M = \alpha K$ ; one clause replaced every  $\tau$  ticks).
- Two learning channels: **private observation** (rate  $p_{\text{obs}}$ ) and **communication** (receiver-pull from in-neighbours), with the per-agent rate normalised across topologies:  $s = 1/\langle k_{\text{in}} \rangle$ .
- Upon learning about a new clause, each agent would try flipping the variables in their private assignments that are involved and keep the change if it reduces the number of known clauses violated (local greedy optimisation)

# Examining the behaviour of information inside the model...

**Observation:** after a number of ticks, the network was filled with mostly stale information.

- Nothing ever removes a clause from a knowledge base: once learned, a clause *outlives its own death* in  $\mathcal{C}$  and keeps being re-transmitted as if new.
- $\sim 70\%$  of held knowledge was stale (i.e. false) by  $t = 2000$  at  $\tau = 10$ ;
- Communication — the only structure-*dependent* channel — was mostly noise; the only reliable truth source was private observation, which is structure-*independent*.

⇒ With 4 in 5 communicated clauses dead on arrival, no topology could have differentiated itself.

**While we couldn't know for sure that this was uniquely driving the result, we do know that - with communicated information comprising of fake news - the model couldn't behave otherwise**

## Tweak 1 — "learning by testing" (signed beliefs)

Replaces passive observation with **active verification**. Each tick, with equal probability, the agent either:

- ① **tests a new clause**: draws a candidate from the universe of possible clauses ( $K(K - 1)$  of them) that it holds no belief about, and keeps it as a *positive belief* iff it actually exists in  $\mathcal{C}$ ; or
  - ② **tests a belief held in its knowledge base** (positive or negative): if they find that a clause does not exist any longer, the clause is kept as a *negative belief* (i.e., "this constraint no longer exists").
- $\Rightarrow$  Knowledge base holds **signed beliefs**: beliefs about a clause existing, and beliefs about a clause not existing.
    - Only positive beliefs drive the greedy optimisation.
    - Size of the knowledge set was doubled to  $2M$  (this is relatively arbitrary)
  - Communication transmits **both polarities**: the receiver adopts the sender's sign on a clause they disagree about.

The idea is to let the network flush fake news.

## Tweak 2 — “communication recency” (transmit the most recently learned belief)

### Recency rule

Upon communication, the sender passes on its most recent belief — most recently learned — among those the receiver doesn't share. Freshest news travels first; old items are forwarded only once everything newer is already shared.

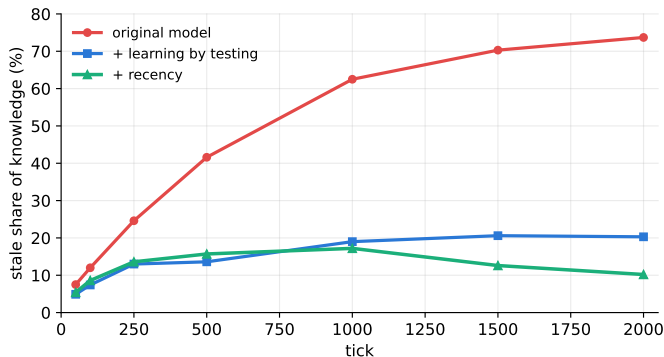
**Effect on information quality** (share of held positives that are stale):

|                            | original model | learning by testing & recency |
|----------------------------|----------------|-------------------------------|
| slow drift ( $\tau = 10$ ) | $\sim 70\%$    | $\sim 10\%$                   |
| fast drift ( $\tau = 1$ )  | $\sim 80\%$    | $\sim 40\text{--}50\%$        |

Stale information now disappears within a few hundred ticks of a clause's death: communication carries mostly *live* content.

Most importantly, network structure likely to play a role, as better connected individuals get good information more quickly

## The tweaks in action: stale information is flushed



In the original model stale knowledge climbs without bound ( $\sim 74\%$  by  $t = 2000$ ). Learning by testing flushes most of it ( $\sim 20\%$ ); transmitting the freshest belief first (recency) roughly halves what remains ( $\sim 10\%$ ).

Scale Free,  $\tau = 10$ , 5-seed means. Each tweak is added on top of the previous.

## New default configuration

**Default configuration** from here on:  $N = 80$ ,  $K = 30$ ,  $\alpha = 2$ ,  $\tau = 1$  (fast drift),  $T = 2000$ , communication normalised, testing + recency on; 10 seeds; long run = last 500 ticks.

Fast drift increases the churn of information, making structure more important (I hope).

# Two hypotheses

## H1 — Structure

Does the *shape* of the network — random, small-world, scale-free, hierarchical — change collective performance?

## H2 — Position

Does an agent's *location* in the network — central vs. peripheral — change its relative performance?

Both are “yes” in organisation theory. We test them with communication volume held equal across networks.

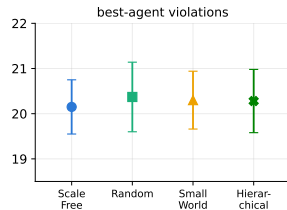
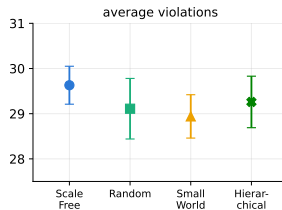
# H1 (structure): **denied**

Four topologies, matched intake ( $\langle k_{in} \rangle \approx 4$ ), 10 seeds. Long-run performance is the same everywhere:

- average violations 28.9–29.6
- best-agent violations 20.2–20.4
- homogeneity 0.69–0.72

Gaps are within seed noise – at least under the normalisation of communication rate. The fixes to the model didn't change this, so independence is not an artefact of stale information. It seems to be a feature of the model.

It's supported by the theory derivations.

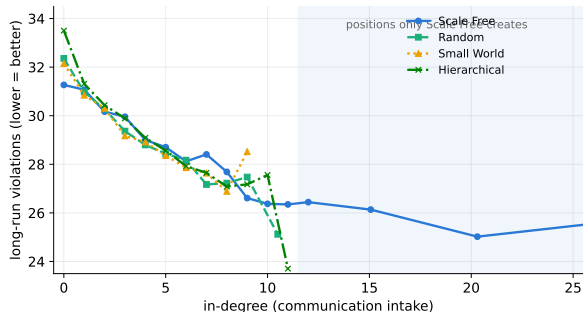


## H2 (position): validated

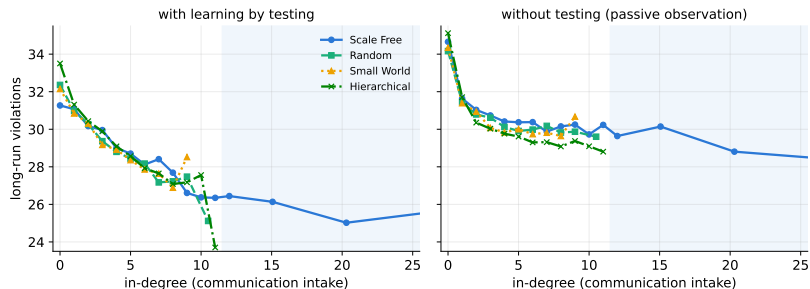
*Population* average performances might be equalised, but each agent's performance scales with its own in-degree.

- central agents have fewer violations in **40/40 runs** ( $\rho \approx -0.5$  to  $-0.6$ ).

**Performance vs. in-degree is one curve, shared by every topology.** Structure only decides which points on it exist.

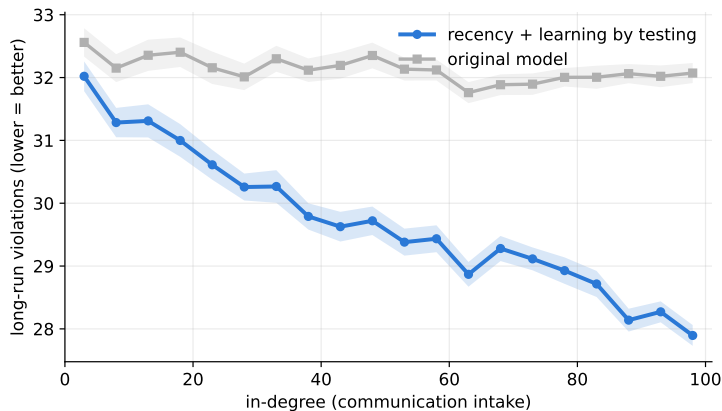


# Why the original model hid this



With testing (left), communication yields *useful* data and the curve keeps falling. Without it (right), most communication is fake news — past the first links, more intake adds nothing. In other words, position pays only once information is trustworthy.

## Position pays across the whole range (designed uniform-degree net)



I believe the milder slope reflects this network's high average degree: under normalisation ( $s = 1/\langle k_{in} \rangle$ ) a given in-degree carries less data. So it's the same curve stretched over a wider axis.

# Implication for organisation design

- Centralisation does not raise average performance: strong hubs are paid for by a larger weak periphery.
- However, network structure can buy **intentionality**. If decision-making or control rights are scarce, one can design organisations centralised around decision makers. Other things being equal, this guarantees better performances (i.e., more accurate problem solving).

# But the theory currently says position shouldn't matter

The degree-resolved mean field (Atiyab):

$$\frac{dp_k}{dt} = R_k [(1 - p_k) \Pr(Z > 0) - p_k \Pr(Z < 0)]$$

At steady state the bracket vanishes, so the rate  $R_k$  cancels: every degree class reaches the same  $p_k^*$ . Position should set only the *speed* of convergence — yet the simulation shows a lasting position gradient.

## Where the gap likely hides

The ODEs let every agent sample the same clause distributions and treat all held clauses as true. But the true binding quantity is *validity*, which rises with intake and is therefore  $k$ -dependent. A validity state  $q_k$  (plus the negative-belief channel) might recover the gradient.